

IMAT MOCK EXAM 10 (Legacy): 2026 Blueprint

60 questions · 100 minutes · scoring: +1.5 correct, -0.4 wrong, 0 blank · personal target: 44/90.0 · Mock exam

Section A - Reading Skills, General Knowledge & Logical Reasoning (Q1-9)

1. Screening programmes aim to detect disease in people who have no symptoms, on the assumption that treating earlier improves outcomes. Yet early detection has a structural drawback. Some of the cancers that screening finds grow so slowly that they would never have caused symptoms in a person's lifetime; counting these cases as cures makes screening appear more effective than it really is, a phenomenon known as overdiagnosis. At the same time, trials show that breast cancer screening does reduce deaths from breast cancer, though the absolute benefit for any individual woman is small, because the disease is uncommon in any single year. Deciding whether to attend screening therefore involves weighing a small reduction in the risk of death against the possibility of unnecessary treatment.

Which of the following statements is most strongly supported by the passage?

- A)** Screening programmes can only succeed against diseases that produce symptoms at an early stage.
- B)** Overdiagnosis occurs because screening detects some cancers that would never have harmed the person.
- C)** The absolute benefit of breast cancer screening is large, because the disease is common.
- D)** Breast cancer screening has been shown to reduce deaths from all causes.
- E)** The passage implies that women should generally be advised not to attend screening.

2. Professional ecologists cannot survey species across entire countries with the staff and funding available to them. Citizen science projects address this gap by recruiting volunteers to record what they observe: birds counted in gardens, butterflies photographed on walks, plants logged along roadsides. The resulting datasets have repeatedly revealed changes in the distribution and timing of species, such as earlier flowering, that smaller professional studies would have missed. The approach has limits, however. Volunteers sample unevenly, recording more near cities and at weekends, and they more easily notice conspicuous species. Project organisers therefore write structured protocols, provide identification training and statistically adjust for sampling bias, treating volunteer data not as a cheap substitute for professional monitoring but as a complement that extends its geographical reach.

Which of the following statements follows from the passage?

- A)** Citizen science projects can extend the geographical coverage of monitoring beyond what professional ecologists could achieve alone.
- B)** Data collected by volunteers are too unreliable to inform ecological research.
- C)** Volunteers overlook conspicuous species because these are difficult to photograph.
- D)** Statistical adjustment eliminates sampling bias from volunteer-collected data.
- E)** Professional ecologists have largely stopped carrying out their own surveys since volunteers took over the work.

3. (General knowledge) The chorus *Va, pensiero*, known in English as the Chorus of the Hebrew Slaves and often treated as an unofficial Italian anthem, comes from the opera *Nabucco*, first performed at La Scala in 1842. Who was the composer?

- A)** Gioachino Rossini
- B)** Giacomo Puccini
- C)** Vincenzo Bellini
- D)** Giuseppe Verdi
- E)** Gaetano Donizetti

4. When Alfred Wegener proposed continental drift in 1912, most geologists dismissed him. He had assembled an impressive case: the fit of the Atlantic coastlines, identical fossils on opposite shores, and traces of ancient glaciation in what are now tropical latitudes. What he lacked was a credible driving force, and his suggestion that continents plough through the ocean floor was physically implausible. The balance changed only in the decades after the Second World War, when mapping of the ocean floor revealed mid-ocean ridges and magnetic stripes recording seafloor spreading, and when the locations of earthquakes traced out the boundaries of moving plates. Drifting continents, once rejected, became the organising principle of modern geology not because Wegener's evidence was rediscovered but because new kinds of evidence, gathered with instruments he never had, settled the question of mechanism.

Which of the following statements is supported by the passage?

- A)** Ocean-floor mapping showed that mid-ocean ridges and magnetic stripes do not exist.
- B)** The question of what drives continental movement remains unanswered in modern geology.
- C)** Plate tectonics won acceptance once Wegener's original evidence was rediscovered after the Second World War.
- D)** Wegener's theory was rejected mainly because the observations behind it were later shown to be mistaken.
- E)** The mechanism Wegener proposed for continental movement was regarded as physically implausible.

5. All the guides at the city museum speak at least three languages. Nobody who speaks at least three languages ever needs an interpreter on a tour. Which one of the following statements must be true?

- A)** Everyone who speaks at least three languages works as a guide at the city museum.
- B)** Every guide at the city museum works full time.
- C)** No guide at the city museum ever needs an interpreter on a tour.
- D)** Some guides at the city museum need an interpreter on some tours.
- E)** Anyone who never needs an interpreter on a tour speaks at least three languages.

6. Five colleagues, Elena, Jonas, Nadia, Omar and Priya, sit around a circular table with five chairs numbered 1 to 5 in clockwise order, one person per chair. Elena sits in chair 1. Nadia sits immediately clockwise from Elena. Omar does not sit next to Elena. Jonas sits immediately anticlockwise from Priya. Which one of the following statements must be true?

- A)** Omar sits next to Elena.
- B)** Nadia and Priya sit next to each other.
- C)** Omar sits between Nadia and Jonas.
- D)** Priya sits immediately clockwise from Elena.
- E)** Jonas sits immediately anticlockwise from Elena.

7. Five coloured balls, one each of blue, green, red, white and yellow, are placed in five boxes numbered 1 to 5 from left to right, one ball per box. The green ball is immediately to the left of the red ball. The red ball is in a box with an even number. The yellow ball is somewhere to the right of the green ball. The white ball is immediately to the right of the yellow ball. The blue ball is in box 5. In which box must the yellow ball be?

- A)** Box 3 **B)** Box 1 **C)** Box 5 **D)** Box 4 **E)** Box 2

8. One term is missing from the following sequence: 5, 8, 14, ?, 35. What is the missing term?

- A)** 18 **B)** 20 **C)** 22 **D)** 23 **E)** 26

9. What is the angle between the hour hand and the minute hand of an analogue clock at 5:30?

- A)** 5 degrees **B)** 15 degrees **C)** 25 degrees **D)** 30 degrees **E)** 45 degrees

Section B - Biology (Q10-32)

- 10.** Glycolysis is often divided into an energy investment stage and an energy payoff stage. Which of the following statements about the breakdown of one molecule of glucose to pyruvate is true?
- A)** ATP is invested to pump pyruvate across the mitochondrial membrane before the payoff stage can begin.
 - B)** Four ATP molecules are consumed in the investment stage and only two are produced in the payoff stage, so glycolysis causes a net loss of ATP.
 - C)** The investment stage can proceed only when oxygen is available to withdraw electrons from the pathway.
 - D)** The net yield is four ATP per glucose, because each ATP forming reaction occurs only once per glucose molecule.
 - E)** The two ATP molecules are spent before the six carbon sugar is cleaved into two three carbon molecules, so every reaction of the payoff stage happens twice for each glucose that enters.
- 11.** In eukaryotic cells, the pyruvate produced by glycolysis does not enter the Krebs cycle directly. Which statement describes what happens to it first?
- A)** It is converted to lactate inside the mitochondrial matrix so that NAD^+ can be regenerated for glycolysis.
 - B)** It crosses both mitochondrial membranes unchanged and condenses directly with oxaloacetate to form citrate.
 - C)** In the mitochondrial matrix it is oxidatively decarboxylated: one carbon leaves as CO_2 , NAD^+ is reduced to NADH , and the remaining two carbon acetyl group is attached to coenzyme A.
 - D)** It is converted to acetyl-CoA in the cytosol, and the acetyl-CoA is then carried into the mitochondria by transport vesicles.
 - E)** It is fully oxidised to CO_2 and water at the inner mitochondrial membrane by the cytochromes of the electron transport chain.
- 12.** Which of the following statements about the Krebs (citric acid) cycle is true?
- A)** Oxaloacetate is consumed in stoichiometric amounts, so the diet must continuously supply it for the cycle to keep turning.
 - B)** For each acetyl group oxidised, one turn of the cycle releases four molecules of CO_2 .
 - C)** Most of the ATP generated in aerobic respiration is produced by substrate level phosphorylation inside the cycle itself.
 - D)** One reaction of the cycle consumes molecular oxygen directly, which is what links the cycle to the electron transport chain.
 - E)** Each turn regenerates the four carbon acceptor oxaloacetate, so the cycle can continue as long as acetyl-CoA, NAD^+ , FAD and ADP are supplied.
- 13.** In the mitochondrial electron transport chain, which complexes pump protons (H^+) from the matrix to the intermembrane space?
- A)** Complexes II and III are the only pumping complexes, because electrons reach both of them at the same energy level.
 - B)** Complexes I, III and IV pump protons, while complex II passes electrons to ubiquinone without pumping any.
 - C)** Only complex IV pumps protons, and it does so while reducing oxygen to water.
 - D)** All four complexes pump protons, which is why NADH and FADH_2 yield exactly the same number of ATP.
 - E)** No complex pumps protons directly; protons cross the membrane together with the electrons inside the mobile carriers.

- 14.** Actively respiring mitochondria maintain a pH difference between the matrix and the intermembrane space. Which statement is correct?
- A)** The intermembrane space is more acidic than the matrix, because electron transport pumps H^+ out of the matrix and the resulting gradient drives H^+ back through ATP synthase.
 - B)** The intermembrane space is more alkaline, because ATP synthesis consumes protons inside the matrix faster than electron transport can replace them.
 - C)** The two compartments have identical pH values, because protons diffuse freely through the outer mitochondrial membrane until equilibrium is reached.
 - D)** No significant pH difference can build up, because the inner mitochondrial membrane is freely permeable to protons.
 - E)** The matrix is more acidic than the intermembrane space, because ATP synthase releases H^+ into the matrix as it converts ADP and phosphate into ATP.
- 15.** During a 400 m sprint, fast muscle fibres convert pyruvate to lactate. Which statement correctly describes the subsequent fate of this lactate?
- A)** It is converted back into glucose inside the same muscle fibre and immediately exported into the blood.
 - B)** It remains locked inside the fibre as a permanent inert waste product.
 - C)** It is excreted unchanged by the kidneys, which greatly lowers the pH of the urine after every sprint.
 - D)** Much of it is carried in the blood to the liver, where it is converted back into glucose that can be returned to the blood or stored as liver glycogen.
 - E)** It is converted by human muscle enzymes into ethanol and CO_2 , exactly as in yeast fermentation.
- 16.** The rate of an enzyme catalysed reaction is measured with and without inhibitor X at a series of substrate concentrations. X lowers the rate at every concentration, but at very high substrate concentration the inhibited reaction approaches the same maximum velocity as the uninhibited reaction. Which conclusion is correct?
- A)** X is a competitive inhibitor: the apparent K_m is raised but V_{max} is unchanged, because sufficiently concentrated substrate outcompetes X for the active site.
 - B)** X binds only to the enzyme substrate complex, so its inhibitory effect becomes stronger as substrate concentration rises.
 - C)** X works by raising the activation energy of the catalysed reaction, so no concentration of substrate can overcome it.
 - D)** X irreversibly destroys the active sites, so adding more substrate can never bring the rate back toward the uninhibited maximum.
 - E)** X is a non-competitive inhibitor that lowers V_{max} , and the apparent convergence of the two curves must be an experimental artefact.
- 17.** The vitamin niacin (vitamin B3) is a precursor of the nicotinamide ring of NAD^+ . Severe dietary deficiency of niacin would be expected to impair most directly the activity of:
- A)** hydrolytic enzymes such as amylase and trypsin, because NAD^+ supplies the water used to break glycosidic and peptide bonds.
 - B)** dehydrogenases of pathways such as glycolysis and the Krebs cycle, because NAD^+ is a coenzyme that accepts electrons from oxidised substrates and must be continuously regenerated.
 - C)** only the protein complexes of the electron transport chain, because NAD^+ is split there to release the energy stored in ATP.
 - D)** the sodium potassium pump, because NAD^+ is hydrolysed to provide the energy for active transport.
 - E)** every enzyme in the body, because NAD^+ is needed for the correct folding of all newly made proteins.

- 18.** According to the sliding filament model, which of the following occurs within a sarcomere during a strong contraction of a skeletal muscle fibre?
- A)** The thick filaments shorten because the myosin tails coil up like springs around the actin filaments.
 - B)** The thin filaments shorten because actin monomers are stripped from their ends during force generation.
 - C)** The H zone widens as the thin filaments are pulled away from the centre of the sarcomere.
 - D)** The thick and thin filaments keep their original lengths while the sarcomere shortens: the I band and the H zone become narrower but the A band keeps its width.
 - E)** The A band and the I band shorten in the same proportion, reflecting the equal shortening of thick and thin filaments.
- 19.** Which of the following statements about motor units in skeletal muscle is true?
- A)** Force can be increased only by raising the firing frequency of a fixed set of motor units, never by recruiting additional units.
 - B)** During a weak contraction, all of the muscle's motor units are active, but each one fires action potentials of reduced amplitude.
 - C)** A motor unit consists of one motor neuron together with all the muscle fibres it innervates, and when that neuron fires, every fibre in the unit contracts.
 - D)** Muscles used for fine, precise movements, such as the external muscles of the eye, are organised into a small number of large motor units.
 - E)** A motor unit consists of a single muscle fibre together with all the motor neurons that supply it.
- 20.** Which of the following correctly matches an epithelial type with its normal location and the advantage it provides there?
- A)** Ciliated pseudostratified columnar epithelium; the walls of the alveoli; the beating cilia stir the air to speed diffusion.
 - B)** Simple squamous epithelium; the lining of the urinary bladder; its many cell layers protect against acidic urine.
 - C)** Transitional epithelium; the lining of the small intestine; its ability to stretch provides a huge absorptive surface.
 - D)** Simple columnar epithelium; the dry outer surface of the skin; its single layer heals wounds faster than any multilayered epithelium.
 - E)** Keratinised stratified squamous epithelium; the epidermis of the skin; its dead, keratin filled surface cells resist abrasion and reduce water loss.
- 21.** Which of the following statements about connective tissues is true?
- A)** Adipose tissue stores fat droplets in a sparse network of collagen fibres rather than inside cells.
 - B)** Cartilage contains no blood vessels: its chondrocytes sit in matrix filled spaces called lacunae and receive nutrients by diffusion, which is one reason injured cartilage heals slowly.
 - C)** The collagen fibres of tendons are secreted by chondrocytes that migrate in from nearby cartilage.
 - D)** Dense regular connective tissue such as tendon is rich in ground substance and poor in fibres, which makes it highly elastic.
 - E)** Blood is not classified as connective tissue because it contains no extracellular matrix.

- 22.** A protein is about to be secreted from a pancreatic cell. Which statement correctly describes its passage through the Golgi apparatus?
- A)** The protein is synthesised on ribosomes attached to the Golgi membrane itself and released directly into its cisternae.
 - B)** The Golgi directs proteins only to lysosomes and the nucleus, while secreted proteins bypass it entirely.
 - C)** Transport vesicles from the endoplasmic reticulum fuse with the cis face, the protein is modified and sorted as it moves through the cisternae, and secretory vesicles bud from the trans face.
 - D)** Vesicles from the endoplasmic reticulum enter the Golgi at the trans face and leave from the cis face once modification is complete.
 - E)** The Golgi returns all processed proteins to the endoplasmic reticulum, which then exports them from the cell.
- 23.** In a gland cell that is actively exporting proteins, which statement about ribosomes is true?
- A)** Each ribosome is permanently bound or permanently free for its whole life, a status fixed when it is assembled in the nucleolus.
 - B)** Bound ribosomes are structurally larger than free ribosomes, and only bound ribosomes can translate mRNA.
 - C)** Secreted proteins are translated inside the Golgi apparatus, so ribosomes take no part in their synthesis.
 - D)** Free ribosomes synthesise all proteins destined for export, which are then imported into the rough ER after they have folded.
 - E)** Free ribosomes and ribosomes bound to the rough ER are structurally the same, and a ribosome binds to the ER because the protein it starts to translate carries a signal sequence.
- 24.** Which of the following statements about the sodium potassium pump (Na^+/K^+ ATPase) of animal cells is true?
- A)** Its net effect is to make the inside of the cell electrically positive with respect to the outside.
 - B)** It moves two Na^+ out of the cell and three K^+ in for each ATP hydrolysed, so it makes the interior slightly positive.
 - C)** It carries Na^+ and K^+ down their concentration gradients and spends ATP only to make the carriers cycle faster.
 - D)** It hydrolyses one ATP for each cycle in which it moves three Na^+ out of the cell and two K^+ in, maintaining gradients that secondary active transport and nerve impulses depend on.
 - E)** It is a ligand gated channel that opens to let Na^+ and K^+ diffuse through whenever ATP binds to it.
- 25.** In blood passing through the capillaries of a vigorously contracting muscle, the local PCO_2 and the H^+ concentration both rise. What effect does this have on haemoglobin?
- A)** Haemoglobin's affinity for oxygen falls, the dissociation curve shifts to the right, and at any given PO_2 a larger fraction of the bound oxygen is unloaded to the muscle.
 - B)** The heat produced by the muscle shifts the curve to the left and thereby cancels out the effect of the extra CO_2 .
 - C)** Haemoglobin's affinity for oxygen rises, the dissociation curve shifts to the left, and oxygen is released only at very low PO_2 values.
 - D)** Affinity is unchanged; the extra CO_2 simply displaces oxygen from the haeme iron in a one for one exchange.
 - E)** The affinity falls because H^+ competes with oxygen for binding to the iron atoms of the haeme groups.

- 26.** A patient spends four months completely immobile in bed. Which statement best describes what happens to bone remodelling in the unloaded bones?
- A)** Bone mass is controlled only by calcium regulating hormones, which are unaffected by loading, so no change occurs.
 - B)** Deposition by osteoblasts rises to compensate for the missing load, so the bones become denser than before.
 - C)** Resorption by osteoclasts exceeds deposition by osteoblasts, so bone mass falls, because mechanical loading is a normal stimulus for bone formation.
 - D)** Osteoclasts are inactivated by disuse, while osteoblasts convert compact bone into spongy bone of equal mass.
 - E)** Osteoblasts and osteoclasts both stop working, so bone mass remains exactly constant.
- 27.** Which of the following statements about the adrenal medulla is true?
- A)** It is regulated by releasing hormones from the hypothalamus, which stimulate the anterior pituitary to act on it.
 - B)** Its hormones slow the heart and promote glycogen storage, preparing the body for rest and digestion.
 - C)** It secretes steroid hormones that diffuse into target cells and bind to nuclear receptors.
 - D)** Its main secretion, cortisol, raises blood glucose during prolonged stress.
 - E)** It is directly innervated by preganglionic sympathetic neurons, so its hormones can be released into the blood within seconds of a stressful stimulus, without any pituitary hormone intervening.
- 28.** Which of the following is a normal function of the human liver?
- A)** Converting the ammonia released by amino acid breakdown into urea, which the blood carries to the kidneys for excretion.
 - B)** Manufacturing red blood cells throughout adult life.
 - C)** Producing insulin and glucagon in response to changing blood glucose.
 - D)** Secreting digestive enzymes into the duodenum through the bile ducts.
 - E)** Storing the bile produced by the gallbladder until fat reaches the small intestine.
- 29.** Which of the following statements about a triglyceride molecule is true?
- A)** In a saturated triglyceride the fatty acid chains contain double bonds that introduce kinks and lower the melting point.
 - B)** It consists of one glycerol joined by ester bonds to three fatty acids, and it is so non polar that it is essentially insoluble in water.
 - C)** Its fatty acids are attached to glycerol by peptide bonds, so dietary fat is digested by proteases.
 - D)** It is built from three glycerol molecules condensed onto a single fatty acid chain.
 - E)** It carries a phosphate group in place of one fatty acid, which is why triglycerides form the bilayers of cell membranes.
- 30.** Which statement correctly describes quiet exhalation at rest?
- A)** Air leaves the lungs because alveolar pressure falls below atmospheric pressure during exhalation.
 - B)** The diaphragm flattens further during exhalation, pressing the abdominal organs upward against the lungs.
 - C)** Quiet exhalation is powered by contraction of the internal intercostal muscles and the abdominal muscles, which squeeze the air out of the lungs.
 - D)** The diaphragm and external intercostal muscles relax, the lungs and chest wall recoil elastically, alveolar pressure rises above atmospheric pressure, and air flows out with no active muscle contraction.
 - E)** Quiet exhalation uses exactly the same muscles as quiet inhalation, but contracts them in the reverse direction.

- 31.** In a plant species, tall stem (T) is dominant over dwarf (t). A tall plant of unknown genotype is crossed with a dwarf plant, and the offspring consist of 61 tall and 59 dwarf plants. What conclusion follows?
- A)** The tall parent is heterozygous (Tt), because crossing a heterozygote with a homozygous recessive plant yields a 1:1 phenotype ratio, reflecting segregation of T and t into equal numbers of gametes.
 - B)** The tall parent is homozygous TT, and the dwarf offspring must have arisen by new mutation.
 - C)** The dwarf parent must actually be heterozygous, since half of the offspring grew tall.
 - D)** The 1:1 ratio indicates incomplete dominance, the heterozygotes being of intermediate height.
 - E)** The genotype of the tall parent cannot be deduced from the phenotypes of the offspring.
- 32.** Most mature neurons of an adult human never divide again, whereas liver cells can resume division after part of the liver is surgically removed. Which statement explains this difference?
- A)** Liver cells regenerate the organ by meiosis, followed by fusion of the cells formed.
 - B)** Neurons remain permanently arrested at the metaphase checkpoint, with chromosomes aligned but unable to separate.
 - C)** Both cell types are outside the cycle, and liver regrowth occurs purely by enlargement of existing cells, with no new cell divisions.
 - D)** Neurons keep replicating their DNA but cannot condense their chromosomes, so they become polyploid.
 - E)** Neurons are held in a permanent G0 phase outside the division cycle, while liver cells remain able to re-enter the cycle from G0 when growth signals arrive.

Section C - Chemistry (Q33-47)

- 33.** Acidified potassium dichromate(VI) solution oxidises iodide ions to iodine: $\text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 14\text{H}^+(\text{aq}) + 6\text{I}^-(\text{aq}) \rightarrow 2\text{Cr}^{3+}(\text{aq}) + 3\text{I}_2(\text{aq}) + 7\text{H}_2\text{O}(\text{l})$. Which of the following statements is correct?
- A)** Chromium is oxidised from +6 to +7, while iodine is reduced from 0 to -1.
 - B)** Iodine is oxidised, its oxidation number rising from -1 to 0, while chromium is reduced from +6 to +3.
 - C)** Neither element changes its oxidation number, because this is not a redox reaction.
 - D)** Iodine is reduced from 0 to -1, while chromium is oxidised from +3 to +6.
 - E)** Iodine is oxidised, its oxidation number rising from -1 to 0, while chromium is reduced from +7 to +3.
- 34.** A coil of copper wire is immersed in a solution of silver nitrate, and the reaction $\text{Cu}(\text{s}) + 2\text{Ag}^+(\text{aq}) \rightarrow \text{Cu}^{2+}(\text{aq}) + 2\text{Ag}(\text{s})$ slowly produces a blue solution and silvery crystals on the wire. Which of the following statements is correct?
- A)** Cu is the oxidising agent because it donates two electrons.
 - B)** Cu^{2+} is the oxidising agent because it carries the larger charge.
 - C)** Ag is the reducing agent because it accepts electrons from the copper.
 - D)** Ag^+ is the oxidising agent because it accepts electrons from copper atoms.
 - E)** NO_3^- is reduced to NO while the silver is deposited.
- 35.** According to VSEPR theory, what are the shape of the ammonia molecule, NH_3 , and the approximate H-N-H bond angle?
- A)** tetrahedral, about 109.5 degrees
 - B)** bent, about 104.5 degrees
 - C)** trigonal pyramidal, about 107 degrees
 - D)** linear, about 180 degrees
 - E)** trigonal planar, about 120 degrees

- 36.** Water (Mr 18) boils at 100 °C, whereas hydrogen sulfide (Mr 34) boils at about -60 °C. Which statement explains this difference?
- A)** Water molecules are held together by ionic bonds between H⁺ and OH⁻ ions.
 - B)** Water molecules attract one another through hydrogen bonds, which are much stronger than the dipole-dipole and London forces present in liquid hydrogen sulfide.
 - C)** Water boils higher because breaking the strong O-H covalent bonds inside its molecules requires more energy than breaking S-H bonds.
 - D)** Hydrogen sulfide molecules form hydrogen bonds with each other, while water molecules cannot.
 - E)** Hydrogen sulfide in fact boils higher than water because it has the larger molar mass.
- 37.** 10 cm³ of a 1.0×10^{-2} mol/dm³ solution of hydrochloric acid, which is fully ionised in water, is diluted with water to 100 cm³. Then 10 cm³ of this diluted solution is further diluted with water to 1000 cm³. What is the pH of the final solution?
- A)** 2 **B)** 3 **C)** 4 **D)** 5 **E)** 9
- 38.** Aqueous ethanoic acid is at equilibrium: $\text{CH}_3\text{COOH}(\text{aq}) \rightleftharpoons \text{H}^+(\text{aq}) + \text{CH}_3\text{COO}^-(\text{aq})$. The solution is diluted with pure water at constant temperature. Which of the following statements is correct?
- A)** The value of K_a decreases in proportion to the dilution.
 - B)** The degree of dissociation decreases, because there are fewer acid molecules per unit volume.
 - C)** The pH falls, because dilution causes more H⁺ ions to be produced.
 - D)** The equilibrium shifts to the left, towards undissociated CH₃COOH.
 - E)** The degree of dissociation increases, but the concentrations of all species fall, so the pH rises.
- 39.** Serine is an alpha-amino acid whose side chain is -CH₂OH. Which set of functional groups is present in a molecule of serine?
- A)** amino, carboxyl and hydroxyl
 - B)** carboxyl, hydroxyl and ester
 - C)** amino, hydroxyl and keto
 - D)** amino, carboxyl and amide
 - E)** amino, carboxyl and thiol
- 40.** But-1-ene and but-2-ene are structural isomers with the molecular formula C₄H₈. Which of the following statements is correct?
- A)** Neither alkene shows cis-trans isomerism because their C=C bonds rotate freely.
 - B)** Both alkenes can exist as cis and trans isomers.
 - C)** Only but-2-ene can exist as cis and trans isomers, because in but-1-ene one carbon of the C=C double bond carries two identical hydrogen atoms.
 - D)** But-2-ene shows cis-trans isomerism because rotation about its C=C double bond is completely free.
 - E)** Only but-1-ene shows cis-trans isomerism, because its double bond is at the end of the carbon chain.
- 41.** Sodium, magnesium and aluminium are consecutive elements in the third period of the periodic table. Which statement is correct?
- A)** Their atomic radii increase from sodium to aluminium because each atom contains more electrons.
 - B)** Sodium has the smallest atomic radius of the three because it has the fewest electrons.
 - C)** Their atomic radii decrease from sodium to aluminium because nuclear charge increases while the outer electrons remain in the same shell.
 - D)** Their first ionisation energies decrease from sodium to aluminium because the atoms become heavier.
 - E)** Aluminium is the least electronegative of the three because metallic character increases across a period.
- 42.** A fixed mass of an ideal gas occupies 6.0 dm³ at 27 °C and 150 kPa. What volume will the same mass of gas occupy at 127 °C and 100 kPa?
- A)** 3.0 dm³ **B)** 6.0 dm³ **C)** 8.0 dm³ **D)** 9.0 dm³ **E)** 12.0 dm³

- 43.** Nitrogen dioxide dimerises: $2\text{NO}_2(\text{g}) \rightleftharpoons \text{N}_2\text{O}_4(\text{g})$, with $\Delta H = -57 \text{ kJ/mol}$ for the forward reaction. An equilibrium mixture of the two gases is heated at constant volume. Which statement is correct?
- A)** The equilibrium shifts towards NO_2 and the value of K_c decreases.
 - B)** The equilibrium shifts towards N_2O_4 and the value of K_c stays the same.
 - C)** The equilibrium shifts towards NO_2 and the value of K_c increases.
 - D)** The equilibrium shifts towards N_2O_4 and the value of K_c increases.
 - E)** The position of equilibrium does not change; only the rates of the forward and reverse reactions increase.
- 44.** Which of the following statements about the bonding in benzene, C_6H_6 , is correct?
- A)** All six C-C bonds in benzene are single bonds, and the sixth valence electron of each carbon moves freely like the electrons of a metal.
 - B)** The six carbon atoms form a planar ring with six delocalised pi electrons spread over the whole ring, making all six C-C bonds equivalent and intermediate in length between single and double bonds.
 - C)** The benzene ring is non-planar, with its pi electrons localised on alternating carbon atoms.
 - D)** Benzene undergoes addition in preference to substitution because addition preserves the delocalised system.
 - E)** Benzene contains three separate $\text{C}=\text{C}$ double bonds alternating with single bonds, which explains why it reacts mainly by addition.
- 45.** For a given reversible reaction, the value of the equilibrium constant K_c is changed by which of the following?
- A)** Increasing the total pressure.
 - B)** Increasing the concentration of a reactant.
 - C)** Adding a catalyst.
 - D)** A change in temperature.
 - E)** Adding an inert gas at constant volume.
- 46.** What is the chemical formula of iron(III) sulfate?
- A)** FeSO_4 **B)** $\text{Fe}(\text{SO}_4)_3$ **C)** $\text{Fe}_2(\text{SO}_4)_3$ **D)** $\text{Fe}_3(\text{SO}_4)_2$ **E)** Fe_2S_3
- 47.** Naturally occurring potassium contains the radioactive isotope potassium-40. How many neutrons are there in one atom of potassium-40? (Potassium has atomic number 19.)
- A)** 19 **B)** 20 **C)** 21 **D)** 40 **E)** 59

Section D - Mathematics (Q48-54)

- 48.** Solve the inequality $\sqrt{2x - 5} < 9$.
- A)** $x \leq 43$ **B)** $x < 43$ **C)** $x \geq 5/2$ **D)** $5/2 \leq x < 43$ **E)** $5/2 < x < 43$
- 49.** Solve the inequality $\cos(x) \leq 1/2$ for $0 \leq x \leq 2\pi$.
- A)** $2\pi/3 \leq x \leq 4\pi/3$
 - B)** $\pi/3 \leq x \leq 5\pi/3$
 - C)** $0 \leq x \leq \pi/3$ or $5\pi/3 \leq x \leq 2\pi$
 - D)** $\pi/3 \leq x \leq 2\pi/3$
 - E)** $\pi/3 < x < 5\pi/3$
- 50.** In a survey of 100 tourists visiting Rome, 62 say they will visit the Colosseum, 55 say they will visit the Vatican Museums, and 17 say they will visit neither of the two. How many of the tourists will visit both attractions?
- A)** 17 **B)** 28 **C)** 34 **D)** 38 **E)** 117
- 51.** The polynomial $P(x) = x^3 + ax + b$ is exactly divisible by $(x - 1)$, and when $P(x)$ is divided by $(x + 2)$ the remainder is 6. What is the value of a ?
- A)** -5 **B)** -3 **C)** 1 **D)** 4 **E)** 5

52. A triangle has perimeter 72 cm and its side lengths are in the ratio 3 : 4 : 5. What is the area of the triangle?

- A)** 108 cm² **B)** 216 cm² **C)** 270 cm² **D)** 360 cm² **E)** 432 cm²

53. Marco is 26 years old and his sister Elisa is 14 years old. After how many years will Marco be exactly one and a half times as old as Elisa?

- A)** 2 **B)** 5 **C)** 10 **D)** 12 **E)** 24

54. Solve the equation $3^{(x+3)} = 9^{(x-1)}$.

- A)** -5 **B)** 1 **C)** 3 **D)** 5 **E)** 7

Section E - Physics (Q55-60)

55. On a straight motorway a car travels at a constant 108 km/h. A truck ahead travels in the same direction at a constant 25 m/s. At one instant the truck is 100 m in front of the car. How long does it take the car to draw level with the truck?

- A)** 3.3 s **B)** 4 s **C)** 10 s **D)** 20 s **E)** 33 s

56. A 60 V battery of negligible internal resistance is connected to an 8 Ω resistor in series with a parallel combination of a 20 Ω resistor and a 30 Ω resistor. What current does the battery supply?

- A)** 2 A **B)** 3 A **C)** 4 A **D)** 5 A **E)** 7.5 A

57. A current is passed through molten potassium chloride held between two electrodes. Which one of the following statements is correct?

- A)** No current can flow, because the ions of a molten salt are not free to move.
B) The positive ions and the negative ions drift in the same direction.
C) The K⁺ ions drift toward the positive electrode and the Cl⁻ ions toward the negative electrode.
D) The conventional current in the liquid is carried only by the negative ions.
E) The direction of the conventional current in the liquid is the direction of drift of the positive ions.

58. The acceleration due to gravity at the surface of a planet of radius R is g. At what height h above the surface has the gravitational acceleration fallen to exactly one quarter of its surface value?

- A)** R **B)** 3R **C)** R/4 **D)** 2R **E)** R/2

59. A 10 kg box is pulled along a horizontal floor by a horizontal force of 60 N. The coefficient of kinetic friction between the box and the floor is 0.40. What is the acceleration of the box? (g = 10 m/s²)

- A)** 0.4 m/s² **B)** 2 m/s² **C)** 5.6 m/s² **D)** 6 m/s² **E)** 20 m/s²

60. A rectangular drain hatch of area 0.5 m² lies flat on the bottom of a swimming pool filled to a depth of 4.0 m with water of density 1000 kg/m³. What force does the water exert on the hatch? (g = 10 m/s²; ignore atmospheric pressure.)

- A)** 2000 N **B)** 4000 N **C)** 20000 N **D)** 40000 N **E)** 80000 N

Answer key

This key refers to the option order printed in this paper. The online app shuffles option order on every attempt, so answer letters inside the app will differ from the ones here.

1	B	2	A	3	D	4	E	5	C
6	C	7	A	8	D	9	B	10	E
11	C	12	E	13	B	14	A	15	D
16	A	17	B	18	D	19	C	20	E
21	B	22	C	23	E	24	D	25	A
26	C	27	E	28	A	29	B	30	D
31	A	32	E	33	B	34	D	35	C
36	B	37	D	38	E	39	A	40	C
41	C	42	E	43	A	44	B	45	D
46	C	47	C	48	D	49	B	50	C
51	A	52	B	53	C	54	D	55	D
56	B	57	E	58	A	59	B	60	C